

Research Progress of Large Models and Generative AI in the Management of Urban Solid Waste

Research Progress of Large Models and Generative AI in the Management of Urban Solid Waste: A Perception-to-Governance Capability Framework

Abstract

Urban solid waste management (USWM) faces mounting pressure from rising waste generation, heterogeneous material streams, and the operational ceiling of conventional, task-specific machine learning, which generalizes poorly beyond narrow, single-modality applications. The recent emergence of large models—large language models, vision-language models, and generative architectures such as generative adversarial networks and diffusion models—offers a qualitatively different paradigm, combining broad transferable knowledge, multi-step reasoning, native multimodality, and, increasingly, tool-using agentic autonomy. Yet the overwhelming majority of existing reviews of artificial intelligence in USWM organize this literature by solid-waste life-cycle stage (generation, collection, sorting/treatment, recovery, disposal), an ontology inherited from earlier task-specific tools that is structurally unable to represent what large models do, because their defining property—level of machine cognition and autonomy—cuts horizontally across every stage. This review introduces and applies an original, technology-native alternative: the Perception-to-Governance (P2G) capability framework, which reclassifies the literature along a single ascending continuum of five tiers—Perceive, Comprehend, Generate, Act, and Govern—each presupposing the capabilities of those below it. Synthesizing evidence tier by tier reveals a steep, previously obscured maturity gradient: dense and consolidating evidence at Tier I (multimodal perception and classification), thinning sharply through Tiers II–III (knowledge-grounded reasoning and constrained generation), and becoming nascent to largely prospective at Tiers IV–V (agentic operational control and AI-mediated governance). The review further consolidates cross-cutting issues invisible to a stage-based reading—reliability and hallucination, domain-data and benchmark scarcity, equity, and an “AI sustainability paradox” whereby the environmental footprint of large models must be weighed against their environmental benefit—and translates the resulting gap analysis into a tier-organized research agenda. The P2G framework offers the field both a sharper diagnostic lens and a roadmap for responsibly advancing generative AI across the full breadth of urban solid waste management.

Keywords

Large language models; generative artificial intelligence; urban solid waste management; vision-language models; waste sorting and classification; agentic AI; AI governance; sustainability paradox

1. Introduction

1.1 The urban solid waste predicament and the ceiling of conventional digitalization

Urban solid waste management (USWM) has become one of the defining environmental-engineering challenges of the urban century. Rapid urbanization, rising per-capita consumption, and increasingly heterogeneous material streams have driven municipal solid waste (MSW) generation upward at a pace that consistently outstrips the adaptive capacity of collection, sorting, treatment, and disposal infrastructure [33,36,76]. The consequences span the full environmental-engineering agenda: greenhouse-gas emissions from landfills and incineration, leachate and microplastic contamination of soil and water, resource loss that undermines circular-economy targets, and acute environmental-justice burdens that fall disproportionately on vulnerable communities. Against this backdrop, the field has, over the past decade, embraced data-driven decision support, and a substantial body of work now documents the application of conventional machine learning (ML) and deep learning (DL) across the waste chain—from generation forecasting and smart-bin monitoring to image-based sorting and process optimization [33,34,35,37].

Yet the prevailing generation of artificial-intelligence (AI) tools in USWM shares a common architectural ceiling. Task-specific ML and DL models are, by construction, narrow: each is trained for a single predictive or classificatory task, on a single modality, against a bespoke labeled dataset, and it generalizes poorly beyond its training distribution [35,37]. A waste-generation regressor cannot read a regulation; a sorting classifier cannot explain its decision to an operator; a route optimizer cannot reason about a novel disruption it never saw in training. As a result, the digitalization of waste systems has remained fragmented into disconnected point solutions, and the human expertise required to integrate them—interpreting outputs, reconciling heterogeneous data, and translating analytics into operational and policy decisions—has remained the binding constraint.

1.2 The generative-AI inflection point

The emergence of large models—foundation models, large language models (LLMs), vision-language models (VLMs), and generative architectures such as generative adversarial networks (GANs) and diffusion models—marks a qualitative break from this paradigm. Built on the Transformer architecture [1] and trained at scale, these models exhibit capabilities that earlier task-specific systems do not: broad transferable knowledge acquired through self-supervised pretraining [3,14,15,16], in-context and few-shot adaptation without task-specific retraining [3], emergent multi-step reasoning [11], reliable instruction-following induced through human-feedback alignment [13], native multimodality that fuses images, text, and structured data [4,6,19,20], and—when coupled to external tools—autonomous, goal-directed agency [12,17,18]. Generative models additionally invert the classical analytics workflow: rather than only mapping inputs to labels, they synthesize new artifacts—plausible images, design scenarios, code, structured reports, and grounded natural-language explanations [8,9,10,21,22].

These capabilities are now diffusing rapidly into environmental science and engineering. ChatGPT-class models have been examined as research and decision-support aids across environmental research workflows [23,24], and the most mature engineering deployments to date cluster in the water and wastewater sector, where domain-adapted LLMs, benchmarks, and conversational operational agents are already being constructed and evaluated [25,26,27,28,29]. In parallel, the foundation-model paradigm is being formalized for environmental and Earth-system science more broadly [30,31,32]. For USWM specifically, the leading edge is visible but uneven: vision-language models are beginning to enable zero- and few-shot waste recognition without bespoke retraining [43,44,45,46], and generative models are being used to augment scarce waste-image datasets [52]. The central premise of this review is that large models do not merely add another tool to the USWM toolbox—they

reconfigure the relationship between human expertise and machine capability across the entire waste system.

1.3 Why the prevailing organizing logic fails for generative AI

The overwhelming majority of existing reviews of AI in USWM—by our reading, well over 90%—are organized along the **solid-waste life-cycle stages**: generation, collection and transport, sorting and treatment, resource recovery, and final disposal [33,34,35,36,37]. This material-flow ontology is intuitive and historically productive: it mirrors the physical movement of waste through engineered infrastructure, and it maps cleanly onto the narrow, single-stage tools that have dominated the field.

Precisely because it is anchored to where waste physically is, however, the life-cycle-stage ontology is structurally unable to represent what large models do. Generative-AI capabilities are defined not by their position in the material flow but by their level of machine cognition and autonomy, and they cut horizontally across every stage. A single multimodal model may simultaneously perceive contamination on a sorting line, retrieve and reason over the governing regulation, generate an operator-facing report, and recommend a control action—one model instantiating one capability that is irreducible to any single life-cycle box. Forcing such systems into stage-based compartments produces three recurring pathologies in the literature: (i) fragmentation, in which one model’s role is dismembered across multiple stage chapters and its cross-cutting nature is lost; (ii) occlusion of emergent properties unique to large models—hallucination and reliability, the considerable carbon, water, and material footprint of the models themselves [70,71,72,73], and the human-AI trust and governance relationship [68,69,75]—none of which belong to any waste stage and all of which are therefore systematically under-treated; and (iii) a misleading impression of maturity, because stage-based coverage rewards the abundant, low-cognition applications (e.g., image classification) and masks the near-total absence of higher-order capabilities such as autonomous agentic operation and AI-mediated governance. A new organizing logic, native to the technology rather than to the conveyor belt, is required.

1.4 A Perception-to-Governance capability framework

To address this gap, we propose and adopt an original organizing framework that reclassifies the entire literature according to the **cognitive-agentic function** a large model performs, rather than the life-cycle stage it touches. We term it the **Perception-to-Governance (P2G) capability continuum**: a five-tier ladder of ascending machine cognition and autonomy.

- **Tier I — Perceive.** Multimodal sensing and characterization of waste: recognizing type, composition, contamination, and quantity from images, spectra, and sensor streams [39–51].
- **Tier II — Comprehend.** Knowledge integration and reasoning: ingesting and semantically interpreting heterogeneous domain knowledge—regulations, technical literature, operational logs—to support question answering and decision support [10,23,25,26].
- **Tier III — Generate.** Synthesis and creation: producing synthetic training data, design and operating scenarios, code, and structured reports, and populating digital twins [8,9,21,22,52].
- **Tier IV — Act.** Agentic optimization and control: planning, tool use, and autonomous or semi-autonomous control of collection, sorting, and treatment systems, including reinforcement-augmented decision-making [12,17,18,58,59].
- **Tier V — Govern.** Human-AI symbiosis and governance: public engagement, behavioral influence, policy co-design, and the management of trust, accountability, equity, and the environmental footprint of AI itself [68,69,70,74,75].

[Figure 1 — placement: end of Section 1.4.] Conceptual schematic of the

Perception-to-Governance (P2G) capability continuum. The figure contrasts the conventional linear life-cycle-stage ontology (a horizontal material-flow arrow: generation → collection → sorting/treatment → recovery → disposal) with the proposed P2G ladder rendered as an orthogonal vertical axis of ascending cognition/autonomy (Perceive → Comprehend → Generate → Act → Govern). Arrows from each tier project across all life-cycle stages, visually emphasizing that capabilities are stage-agnostic. A shaded “maturity gradient” indicates dense literature at lower tiers and sparse coverage at upper tiers.

The framework is deliberately technology-native and is intended to do analytical work beyond mere classification. Because the tiers are ordered by cognition and autonomy, mapping each reviewed study onto a tier yields a **maturity-and-gap map** of the field: it quantifies where research clusters (Tier I perception, which is comparatively saturated) and where it thins to near-absence (Tiers IV-V), thereby converting a descriptive survey into a forward-looking research agenda (developed in Sections 8–9).

1.5 Objectives, scope, and contributions

This review synthesizes the state of research on large models and generative AI in urban solid waste management, with municipal solid waste as the primary focus and adjacent streams (construction-and-demolition and electronic waste [78]) included where they illuminate transferable capabilities. We scope “large models and generative AI” to encompass foundation models, LLMs, VLMs and multimodal LLMs, and generative architectures (GANs, diffusion, and Transformer-based generators), while engaging conventional ML/DL as the comparative baseline against which the generative paradigm is assessed. The principal contributions are fourfold:

1. We introduce the **Perception-to-Governance capability framework**, an original, technology-native organizing logic that replaces the dominant life-cycle-stage ontology and resolves its fragmentation, occlusion, and false-maturity pathologies.
2. We provide a **tier-by-tier synthesis** of the literature (Sections 3–7), establishing what generative AI can currently do at each level of cognition and autonomy in USWM, and with what evidence.
3. We deliver a **maturity-and-gap analysis** (Section 8) that exposes the field’s concentration at lower tiers and confronts the cross-cutting issues the life-cycle ontology obscures—reliability and hallucination, data scarcity, standardization, equity, and the “AI sustainability paradox” whereby the environmental cost of large models must be weighed against their environmental benefits.
4. We articulate a **structured research agenda** (Section 9) organized by capability tier, identifying the benchmarks, datasets, model architectures, and governance instruments required to advance the field responsibly.

1.6 Organization of the paper

The remainder of the paper is organized around the P2G framework. Section 2 formally defines the framework and details the review methodology, including the literature search protocol and scientometric analysis. Sections 3 through 7 synthesize the evidence tier by tier—Perceive, Comprehend, Generate, Act, and Govern, respectively. Section 8 presents the cross-cutting maturity-and-gap analysis and critical appraisal. Section 9 sets out the challenges and future research directions, and Section 10 concludes.

[Table 1 — placement: Section 2.] Definitions of the five P2G capability tiers, each row specifying the tier, its defining cognitive-agentic function, representative generative-AI techniques, illustrative USWM tasks, and current maturity level. (Detailed and populated in Chapter/Section 2.)

2. Conceptual Framework and Review Methodology

This section establishes the analytical foundation of the review. Section 2.1 formally defines the Perception-to-Governance (P2G) capability framework and its ordering principle; Section 2.2 specifies the protocol by which individual studies are assigned to capability tiers; and Section 2.3 describes the literature search, screening, and scientometric synthesis on which the subsequent chapters rest.

2.1 The Perception-to-Governance framework: formal definition

The P2G framework reclassifies the literature on large models in urban solid waste management (USWM) according to a single ordering principle: the **level of machine cognition and autonomy** that a generative-AI system exercises in a given application. This principle is deliberately orthogonal to the material-flow ontology that organizes the conventional literature. Where the life-cycle-stage view asks where in the waste stream the technology operates, the P2G view asks what cognitive-agentic function the technology performs—a distinction that, as argued in Section 1.3, is the one that actually discriminates among large-model applications, because a single foundation model can be deployed at any stage of the material flow while remaining fixed in its cognitive role.

We define the continuum as five ordered tiers. The ordering is monotonic in two coupled dimensions—**cognitive sophistication** (from low-level pattern recognition to high-level reasoning and value judgment) and **operational autonomy** (from human-in-the-loop perception to human-AI co-governance of socio-technical systems)—and each tier presupposes and builds upon the capabilities of those below it.

- **Tier I — Perceive.** The system extracts structured meaning from raw, often multimodal, sensory input: identifying waste category, material composition, contamination, defect, or quantity from images, hyperspectral or near-infrared signals, and sensor streams. The cognitive demand is recognition and discrimination; autonomy is minimal, with the human retaining all interpretive and decision authority [39–51].
- **Tier II — Comprehend.** The system integrates and reasons over heterogeneous knowledge rather than raw sensor data—technical literature, regulations, standards, operational records—to answer questions, summarize, and support human decisions. The cognitive demand rises to semantic understanding and contextual reasoning; autonomy remains advisory [10,11,23,25,26].
- **Tier III — Generate.** The system synthesizes novel artifacts: realistic synthetic training data for downstream models, design and operating scenarios, code, structured technical reports, and content that populates simulation environments and digital twins. The defining demand is generative creation under constraints; autonomy is bounded by human specification and acceptance [8,9,21,22,52].
- **Tier IV — Act.** The system plans and executes sequences of actions toward a goal, typically by orchestrating tools, models, and controllers—closing the loop from perception and reasoning to optimization and control of collection, sorting, and treatment operations, with the human supervising rather than directing each step. The defining demand is agentic decision-making and tool use [12,17,18,58,59].
- **Tier V — Govern.** The system participates in the social and institutional regulation of the waste system itself: mediating public engagement and behavior, supporting policy co-design, and surfacing questions of trust, accountability, equity, and the environmental footprint of the AI. The defining demand is value-laden judgment and human-AI symbiosis; “autonomy” here is reframed as shared agency between machines, citizens, and institutions [68,69,70,74,75].

[Table 1 — placement: here, Section 2.1.] Definitions of the five Perception-to-Governance capability tiers. Columns: Tier; Defining cognitive-agentic function; Representative large-model techniques; Illustrative USWM tasks; Current maturity.

Tier	Defining function	Representative techniques	Illustrative USWM tasks	Maturity
I — Perceive	Recognition and discrimination from multimodal sensory input	CNNs, Vision Transformers, YOLO detectors, CLIP/GPT-4V vision-language models, NIR + DL	Waste-stream classification, contaminant and object detection, composition and fill-level estimation	Established / consolidating
II — Comprehend	Semantic understanding and reasoning over domain knowledge	LLMs, retrieval-augmented generation, chain-of-thought prompting, knowledge graphs	Regulation and standard interpretation, technical Q&A, decision support, literature synthesis	Emerging
III — Generate	Constrained synthesis of novel artifacts	GANs, diffusion models, Transformer generators, LLM code/report generation	Synthetic dataset augmentation, scenario and process design, automated reporting, digital-twin content	Early / exploratory
IV — Act	Agentic planning, tool use, and control	LLM agents (e.g., ReAct-style), reinforcement learning, tool-augmented planners	Autonomous routing and dispatch, closed-loop sorting/process control, adaptive operations	Nascent
V — Govern	Value-laden judgment and human-AI co-governance	Conversational agents, behavioral and participatory interfaces, alignment and policy instruments	Citizen engagement and nudging, policy co-design, trust/equity/footprint governance	Largely prospective

The maturity column previews the central empirical finding of the review (Section 8): the literature is dense and consolidating at Tier I, thins progressively through Tiers II-III, and becomes nascent-to-prospective at Tiers IV-V. This gradient is invisible under a life-cycle-stage organization, which distributes the abundant perception work across every stage and thereby conveys a false impression of uniform maturity.

2.2 Tier-assignment protocol

Because large-model studies frequently span more than one tier, an explicit assignment rule is required for reproducibility. We assign each study to the **highest tier at which it makes a substantive, evaluated contribution**, on the rationale that the highest exercised capability is the most informative descriptor of a system’s cognitive-agentic role and the most discriminating for a maturity analysis. A study that uses a generative model only to augment data for a perception classifier, for example, is assigned to Tier I (its evaluated contribution is perception), whereas a study whose central, evaluated object is the synthetic-generation method itself is assigned to Tier III. Studies that genuinely contribute at multiple tiers—an increasingly common pattern as multimodal agents mature—are cross-referenced across the relevant chapters and explicitly flagged as multi-tier in the synthesis, since such studies are themselves a signal of the field’s progression up the continuum. This protocol governs the organization of Sections 3–7.

2.3 Literature search and scientometric synthesis

Search strategy and sources. The review draws on the major bibliographic databases for environmental engineering and computer science—including Scopus, Web of Science, ScienceDirect, IEEE Xplore, SpringerLink, and Wiley Online Library—supplemented by the arXiv preprint repository to capture the fast-moving foundational and methodological literature on large models, which frequently appears there before formal publication. Search strings combined three concept blocks with Boolean operators: a large-model/generative-AI block (e.g., “large language model,” “LLM,” “foundation model,” “generative AI,” “vision-language model,” “diffusion model,” “ChatGPT,” “GPT-4,” “LLM agent”); a waste/environmental block (e.g., “solid waste,” “municipal solid waste,” “waste management,” “recycling,” “waste sorting,” “waste-to-energy,” “circular economy”); and, for capability-specific sweeps, a function block aligned to the five P2G tiers (e.g., “classification/detection,” “knowledge/reasoning,” “synthetic data/generation,” “agent/optimization/control,” “governance/engagement/policy”). Reference lists of the retrieved reviews and key primary studies were back-screened to surface additional sources.

Inclusion and exclusion criteria. We included peer-reviewed journal articles, archival conference papers, and substantive preprints that (i) apply or critically examine a large model or generative-AI method, and (ii) address an urban solid waste task or a directly transferable environmental-engineering task. Given the recency of the field, the temporal emphasis is on 2020–2025, with seminal pre-2020 works on foundational architectures retained for technical grounding. We excluded studies confined to conventional task-specific ML/DL with no generative or large-model component (retained instead only as the comparative baseline of Section 1.1), as well as non-technical commentary lacking methodological or empirical content. Where peer-reviewed evidence in USWM was sparse—as it is for the upper tiers—we incorporated the most mature adjacent evidence, principally from the water and wastewater sector, explicitly labeled as transferable rather than established.

Synthesis approach. Screening and synthesis followed the reporting principles of the PRISMA framework for transparency and reproducibility, adapted to a capability-mapped rather than stage-mapped synthesis; the systematic ML/DL reviews that anchor the conventional literature provide methodological precedent for this design [33,34]. Consistent with the qualitative orientation of this review, we report the screening process and the structure of the evidence base narratively rather than as a quantified flow, reserving exact corpus counts for a future quantitative scientometric study.

[Figure 2 — placement: Section 2.3.] PRISMA-style flow diagram of the literature identification, screening, eligibility, and inclusion process, presented schematically to convey the qualitative pipeline (identification across databases and arXiv → duplicate removal → title/abstract screening → full-text eligibility against the inclusion criteria → tier assignment), without numerical counts.

Scientometric overview. Read qualitatively, the corpus exhibits several robust structural features that motivate the chapters that follow. First, the field is genuinely nascent and sharply accelerating: while deep-learning applications to waste perception predate it, the application of large models and generative AI to USWM is overwhelmingly a post-2020 phenomenon, with a pronounced inflection from 2023 onward that tracks the public release and rapid uptake of ChatGPT-class models [23,24]. Second, the evidence is strongly stratified by capability tier in the manner anticipated above—concentrated in Tier I perception, where vision and, increasingly, vision-language models dominate [39–51], and progressively sparser at higher tiers. Third, the most operationally mature engineering deployments of LLMs in the broader environmental domain are concentrated in the water and wastewater sector—domain-adapted models, benchmarks, and conversational operational agents [25,26,27,28,29]—which therefore serves throughout this review as the leading indicator for what Tiers II–V may become in USWM. Fourth, the upper tiers reveal conspicuous white

space: peer-reviewed work on autonomous LLM agents for waste operations (Tier IV) and on AI-mediated governance of waste systems (Tier V) is scarce to absent, even as the enabling general-purpose methods mature rapidly [12,17,18]. These four features are developed and evidenced tier by tier in Sections 3–7 and consolidated into the maturity-and-gap analysis of Section 8.

[Figure 3 — placement: Section 2.3.] Scientometric overview of the corpus, a multi-panel figure comprising (a) a schematic timeline of annual publication activity highlighting the post-2023 inflection; (b) a qualitative bar representation of evidence density across the five P2G tiers; and (c) a keyword co-occurrence map distinguishing the consolidated perception cluster from the emerging knowledge-reasoning, generation, and governance clusters. Panels are rendered to communicate relative structure and trend rather than precise counts.

3. Tier I — Perceive: Multimodal Sensing and Characterization of Waste

Perception is the foundational tier of the P2G continuum and, by a wide margin, the most mature application of advanced AI in urban solid waste management (USWM). At this tier the system extracts structured meaning—category, material, contamination, defect, fill level, or quantity—from raw sensory input, while the human retains all interpretive and decision authority. This chapter synthesizes the perception literature along the trajectory by which it has climbed toward the “large-model” frontier: from supervised convolutional and Transformer-based recognition (Section 3.1), through real-time detection and its coupling to robotic sorting (Section 3.2), to the qualitatively new capabilities introduced by foundation and vision-language models—open-vocabulary, zero- and few-shot recognition and promptable segmentation (Section 3.3)—and finally to perception beyond the visible spectrum (Section 3.4). Section 3.5 appraises the tier as a whole and motivates why, despite its density, the perception literature has only recently begun to be “large-model” in character.

3.1 From convolutional classification to Vision Transformers

The first wave of deep-learning perception in USWM framed waste characterization as a supervised image-classification problem. Convolutional neural networks (CNNs), typically transferred from ImageNet-pretrained backbones such as ResNet, were fine-tuned to discriminate a small fixed taxonomy of waste categories, and they rapidly displaced hand-engineered features as the standard approach to material recognition on sorting lines and in smart bins [33,35]. Representative of this consolidated paradigm, MSWNet adapts a ResNet-50 backbone through transfer learning to municipal solid waste (MSW) sorting, illustrating both the strengths of the approach—high accuracy on curated category sets—and its characteristic limitations: dependence on large bespoke labeled datasets and brittleness to the visual heterogeneity of real waste streams [40b].

The importation of the Transformer architecture into vision [1,7] extended rather than overturned this paradigm. Vision Transformers (ViTs) replace convolution’s local inductive bias with global self-attention, and hybrid designs that combine convolutional feature extraction with Transformer encoders have been reported to improve waste-classification accuracy over pure-CNN baselines, particularly for visually similar materials where long-range context aids discrimination [49]. Crucially, however, both CNN and ViT classifiers in this wave remain closed-set and task-specific: they recognize only the categories present in their training labels, generalize poorly across facilities and imaging conditions, and cannot accommodate a novel material without retraining. These are precisely the constraints that the foundation-model paradigm later relaxes (Section 3.3), and they are the perception-tier manifestation of the architectural ceiling described in Section 1.1.

3.2 Real-time detection and the perception-action coupling in robotic sorting

Where classification answers “what is this image,” detection answers “what objects are where,” and it is the perception primitive that matters most for automated sorting, where multiple items must be localized in a cluttered scene at conveyor speed. The single-stage YOLO family has become the workhorse here, and a steady cadence of waste-specific variants documents the field’s incremental optimization for accuracy under real-time constraints: MRS-YOLO introduces architectural modifications for high-precision multi-category waste detection [41], and YOLOv12-based models report real-time detection-and-classification performance suitable for embedded deployment [42]. The release of curated, realistically annotated benchmarks has been a decisive enabler of this progress: TACO provides trash annotations of litter in unconstrained (“in context”) environments [39], and the WaRP dataset supplies hierarchically labeled images from operating recycling plants together with weakly supervised segmentation, narrowing the gap between benchmark and plant-floor conditions [40].

Detection is also the tier at which perception begins to couple to action, foreshadowing Tier IV. In robotic sorting, a vision module localizes and classifies items and hands pose information to a pick-and-place controller; perception quality directly bounds sorting throughput and purity. This perception-action coupling is the practical reason that data scarcity at the perception tier became a research bottleneck, and it motivated the generative-augmentation work treated separately under Tier III (e.g., WasteGAN synthesizing training images and segmentation masks to improve downstream sorting perception [52])—a study whose evaluated contribution is generation and which is therefore assigned to Tier III under our protocol (Section 2.2), even though its purpose is to serve perception.

3.3 The foundation-model inflection: vision-language, open-vocabulary, and promptable perception

The developments above, though built on Transformers, remain narrow supervised models. The genuinely large-model shift in waste perception is the arrival of vision-language models (VLMs)—CLIP and its successors, and instruction-tuned multimodal LLMs such as GPT-4V [4,6]—which align images and natural-language descriptions in a shared embedding space and thereby enable **open-vocabulary** recognition: classifying waste against category names supplied as text at inference time, with no category-specific retraining. This directly relaxes the closed-set ceiling of Section 3.1 and is the perception-tier instantiation of the inflection point argued in Section 1.2.

A small but rapidly growing and methodologically coherent body of work now evidences this shift in USWM. Malla et al. demonstrate that prompt engineering over a VLM yields a scalable waste-recognition solution that adapts to new categories through prompt design rather than data collection [43]; Ranjbar et al. report zero- and few-shot multimodal classification of plastic waste, showing that VLMs can recognize material classes from a handful of examples or none at all [44]; and in the adjacent construction-and-demolition stream, vision and language models have been probed for construction-waste material recognition, with domain-adapted contrastive models (ConCLIP) outperforming generic baselines on facility imagery [45]. The maturation of the subfield is signaled by the appearance of dedicated evaluation infrastructure: Waste-Bench benchmarks large vision-language models specifically in cluttered, realistic waste environments, exposing the robustness gap between curated and operational conditions [46].

A parallel advance is **promptable segmentation**. The Segment Anything paradigm, adapted to the domain as WasteSAM, delivers accurate instance segmentation of construction-waste imagery to support recycling, replacing per-dataset segmentation training with a promptable foundation model [47]. Reviews of computer vision in mixed construction-and-demolition material-recovery facilities document the same overall arc—from bespoke CNNs toward trans-

ferable, foundation-model-based perception—while cataloguing the persistent challenges of occlusion, material ambiguity, and dataset scarcity that operational deployment still faces [48].

[Table 2 — placement: here, Section 3.3.] Representative Tier I (Perceive) studies in USWM. Columns: Study (ref.); Waste stream (MSW / plastics / C&D / litter / e-waste); Perception task (classification / detection / segmentation / spectral characterization); Model class (CNN / ViT / YOLO / VLM / SAM / NIR+DL); Learning regime (supervised closed-set / open-vocabulary / zero- or few-shot); Dataset or benchmark; Key reported capability. Rows ordered to trace the progression from closed-set CNN classification → real-time detection → VLM open-vocabulary and promptable segmentation → spectral sensing, visually substantiating the “established / consolidating” maturity rating of Table 1.

3.4 Perception beyond the visible spectrum

Vision alone cannot resolve materials that are visually identical but chemically distinct—most importantly the polymer families within the plastic stream—so a complementary perception literature couples deep learning to spectral sensing. Near-infrared (NIR) and multispectral signatures, long used in industrial sorting, are increasingly paired with neural classifiers: a low-cost multispectral-NIR sensor combined with deep learning has been shown to recognize plastic-waste polymer types at a fraction of the cost of conventional hyperspectral rigs, pointing toward distributed, affordable material identification [50]. Alongside spectral sensing, lightweight detectors engineered for real-time regional and domestic waste recognition demonstrate that perception can be pushed to the edge, onto embedded and mobile hardware close to the point of collection [51]. Together these strands extend Tier I beyond the camera, but they remain, for now, predominantly task-specific deep learning rather than foundation-model perception—an asymmetry to which we return below.

3.5 Appraisal of the Perceive tier

Read as a whole, Tier I is the only tier of the P2G continuum that can be described as established and consolidating. Several structural observations follow, each setting up later chapters. First, the density is real but stratified by capability within the tier: closed-set CNN/ViT classification and YOLO detection are mature and incrementally optimized, whereas open-vocabulary and zero-/few-shot VLM perception—the truly large-model capability—is recent (concentrated in 2024–2025), small in volume, and still establishing its robustness on operational, cluttered, and out-of-distribution waste [43,44,45,46,48]. The tier’s apparent saturation is therefore largely a saturation of pre-foundation-model methods.

Second, the binding constraints have shifted. For closed-set models the constraint was labeled-data scarcity, which drove the perception-generation coupling of Section 3.2; for VLMs the constraint is no longer category coverage but reliability under realistic conditions—occlusion, contamination, domain shift between facilities, and the hallucination and miscalibration that VLMs inherit from their language backbones—exactly the emergent, cross-cutting failure modes that a life-cycle ontology occludes and that we consolidate in Section 8. Third, modality coverage is uneven: the foundation-model wave is overwhelmingly visual, while spectral, acoustic, and other non-visual perception remains task-specific, marking an obvious frontier for multimodal foundation models. Finally, and most consequentially for the argument of this review, the comparative abundance of Tier I makes the thinning of the tiers above it unmistakable: perception is where large-model USWM research currently lives, and the chapters that follow trace how sharply the evidence attenuates as we ascend from perceiving the waste stream to comprehending, generating within, acting upon, and ultimately governing it.

[Figure 4 — placement: end of Section 3.] Schematic of the Tier I perception stack and its evolution. A layered diagram showing, left to right, the progression from (a) closed-set CNN/ViT classifiers and (b) real-time YOLO detectors feeding robotic sorting, to (c) foundation-model perception—VLM open-vocabulary / zero-shot recognition and SAM-style promptable segmentation—and (d) beyond-visible spectral (NIR/multispectral) and edge-deployed sensing. A horizontal “generalization” axis beneath the stack indicates the shift from bespoke-data, closed-set models on the left to prompt-driven, open-vocabulary foundation models on the right; a callout marks data scarcity and operational reliability as the binding constraints handed off, respectively, to Tier III (generation) and the cross-cutting analysis of Section 8.

4. Tier II — Comprehend: Knowledge Integration and Reasoning

Where Tier I extracts meaning from raw sensory signals, Tier II operates on knowledge: the heterogeneous textual and structured information that surrounds a waste system—technical literature, regulations and standards, permits, operational records, and accumulated engineering expertise. At this tier a large language model (LLM) ingests, semantically interprets, and reasons over this corpus to answer questions, summarize, and support human decisions, while autonomy remains strictly advisory—the model informs a human who decides and acts. This is the first tier at which the defining technology is the language model rather than the vision model, and it marks the transition from seeing the waste stream to understanding the body of knowledge that governs its management. This chapter sets out the enabling capabilities and their grounding methods (Section 4.1), surveys the still-thin direct evidence in USWM together with the mature leading-indicator evidence from the water sector (Sections 4.2–4.3), and appraises the tier’s emerging status and its characteristic risk—fluent unreliability (Section 4.4).

4.1 Enabling capabilities: in-context reasoning and retrieval grounding

Three properties of large language models make Tier II possible, none of which is available to the task-specific models of Tier I. First, broad pretrained knowledge and **in-context, few-shot adaptation** allow a single model to address heterogeneous queries—from “which polymer codes are accepted in this kerbside scheme” to “summarize the abatement obligations in this incineration permit”—without task-specific retraining [3,4]. Second, **multi-step reasoning**, elicited through chain-of-thought and related prompting strategies, lets the model decompose a multi-condition regulatory or technical question into intermediate inferences rather than pattern-matching a single answer [11]. Third, and decisively for an engineering domain in which fabricated content is unacceptable, **retrieval-augmented generation (RAG)** couples the model to an external, authoritative document store, retrieving relevant passages at inference time and conditioning the response on them [10]. RAG is the central architectural motif of this tier because it directly targets the failure mode that would otherwise disqualify LLMs from regulatory and operational use—hallucination—by grounding outputs in verifiable source text and enabling citation back to the governing document. Knowledge graphs play a complementary structuring role, encoding the entities and relations of a waste-management domain so that retrieval and reasoning can be made more precise and auditable.

4.2 Direct evidence in USWM: an emerging and uneven frontier

Direct, peer-reviewed application of LLM-based comprehension to USWM tasks is, at present, sparse—a thinness that is itself a finding of this review and a direct consequence of the field’s recency (Section 2.3). The general case for LLMs as research and decision-support aids in environmental science has been made and critically examined: ChatGPT-class models have

been assessed for environmental research workflows—literature synthesis, drafting, hypothesis support, and data interpretation—with explicit attention to both their promise and their reliability limits [23,24]. These appraisals establish the plausibility and the perils of Tier II in the environmental domain generally, but they stop short of waste-specific, evaluated deployments.

Within USWM specifically, comprehension capability currently surfaces less as standalone “ask-the-regulation” systems than as the language-and-reasoning component of multimodal systems that also perceive. The vision-language work of Tier I is the clearest example: prompt-engineered and zero-/few-shot waste recognition [43,44] relies on the model’s comprehension of natural-language category descriptions and contextual cues, blurring the Tier I–Tier II boundary and exemplifying the multi-tier studies anticipated by our assignment protocol (Section 2.2). What remains conspicuously underdeveloped is the dedicated Tier II application: an LLM system that interprets waste regulations and technical standards, answers operator and compliance questions against an authoritative corpus, or synthesizes the heterogeneous records of a waste facility into decision-ready summaries. The enabling methods are mature and general; their rigorous, benchmarked instantiation for waste knowledge is largely absent.

4.3 The water and wastewater sector as leading indicator

The most informative evidence for what Tier II will become in USWM comes not from waste but from the adjacent water and wastewater sector, where engineering deployment of LLMs is markedly more advanced and serves throughout this review as a leading indicator (Section 2.3). The pattern there is instructive because it traces the full Tier II maturation arc that USWM has yet to climb. Early position pieces argued the case for ChatGPT in water-resource management, research, and policy [25,29]. The field then moved decisively beyond generic models toward **domain adaptation and rigorous evaluation**: domain-adapted LLMs for water and wastewater management have been constructed together with curated datasets and benchmarks, directly confronting the question of whether a general model can be made reliable on specialized engineering knowledge [26]; and water-knowledge benchmarks have been built to quantitatively evaluate and advance LLM performance on engineering and research tasks, rather than asserting capability anecdotally [27]. Finally, the sector has produced deployed-facing artifacts: **conversational agents** for water-quality education, communication, and operations, demonstrating that grounded LLM interfaces can be placed in front of operators and the public [28].

This trajectory—from generic enthusiasm, through domain adaptation and benchmarking, to operational conversational agents—is precisely the path that waste-knowledge comprehension has not yet traversed. The transferability is strong: both are utility-scale environmental-engineering domains governed by dense regulation, rich in heterogeneous operational data, and reliant on scarce human expertise. The water sector therefore furnishes both a template for USWM Tier II development and concrete evidence that the binding obstacles are surmountable, while also foreshadowing the requirement—benchmarks and domain adaptation—without which waste-domain LLMs will remain unvalidated.

[Table 3 — placement: here, Section 4.3.] Tier II (Comprehend) evidence: direct USWM applications versus transferable water-sector exemplars. Columns: Study (ref.); Domain (solid waste / water & wastewater / environmental-general); Comprehension task (regulation/standard interpretation, technical Q&A, decision support, literature synthesis, conversational operations); Method (prompting / chain-of-thought / RAG / domain-adapted LLM / benchmark); Maturity of evaluation (position/qualitative → benchmarked → deployed agent). The table is designed to make visible the gap between the abundant, benchmarked water-sector evidence and the sparse, mostly embedded waste-sector evidence, substantiating

the “emerging” maturity rating of Table 1 and the leading-indicator argument of Section 2.3.

4.4 Appraisal of the Comprehend tier

Tier II is best characterized as emerging: the enabling methods (in-context reasoning, chain-of-thought, RAG, knowledge graphs) are mature and general-purpose, the adjacent water sector has demonstrated their engineering viability, yet waste-specific, benchmarked applications remain scarce. Three observations frame the chapters and analysis that follow.

First, the defining risk of this tier is qualitatively different from Tier I’s. A perception model fails visibly—a misclassified item—whereas a comprehension model fails fluently, producing confident, well-formed, and incorrect statements about a regulation or a procedure. In a compliance-critical engineering domain this hallucination risk is not a peripheral caveat but the central design constraint, which is why retrieval grounding and citation-to-source are prerequisites rather than enhancements, and why the water sector’s pivot to benchmarking is the necessary precedent for waste [26,27]. This emergent failure mode is a cross-cutting concern consolidated in Section 8.

Second, the absence of waste-domain benchmarks and curated knowledge corpora is the immediate bottleneck. The water sector advanced precisely by building these [26,27]; USWM lacks equivalents, and constructing authoritative, machine-readable corpora of waste regulations, standards, and operational records—against which RAG systems can be grounded and evaluated—is the single most actionable near-term Tier II research priority (developed in Section 9).

Third, comprehension is the pivot of the entire P2G continuum. It consumes the structured outputs of perception (Tier I) and supplies the reasoning substrate on which generation (Tier III) and, especially, agentic action (Tier IV) depend: an autonomous waste-operations agent is, at its core, a comprehension system coupled to tools and a control loop. The thinness of dedicated Tier II work in USWM therefore both reflects and constrains the still-greater scarcity at the tiers above it, to which the review now turns.

[Figure 5 — placement: end of Section 4.] Schematic of the Tier II comprehension architecture and the USWM-water maturity gap. The left panel diagrams a grounded comprehension pipeline: heterogeneous knowledge sources (regulations, standards, technical literature, operational records) → retrieval-augmented generation over an authoritative document store and domain knowledge graph → reasoning (chain-of-thought) → human-facing outputs (regulatory Q&A, decision support, synthesis) with citation-to-source, the human retaining decision authority. The right panel plots a maturity ladder—position/qualitative → domain-adapted → benchmarked → deployed conversational agent—showing the water/wastewater sector occupying the upper rungs while USWM clusters near the bottom, visually encoding the leading-indicator relationship.

5. Tier III — Generate: Synthesis and Creation

The first two tiers are fundamentally discriminative: Tier I maps sensory input to categories, and Tier II maps queries to grounded answers. Tier III inverts the workflow. Rather than mapping inputs to labels, a generative model synthesizes novel artifacts—plausible images, design and operating scenarios, code, and structured technical reports—under specified constraints, while the human supplies the specification and retains acceptance authority. This is the capability that most sharply distinguishes generative AI from the conventional analytics paradigm of Section 1.1, and yet in USWM it is, at present, early and exploratory: the technique base is mature and general, but its waste-domain application is concentrated almost entirely in a single, instrumental use—synthesizing training data to feed the data-hungry perception models

of Tier I. This chapter distinguishes the generative families and their constraint mechanisms (Section 5.1), examines the dominant data-augmentation application that is the tier’s principal direct evidence (Section 5.2), and then maps the largely prospective frontier—scenario and process design, automated reporting, and digital-twin content—where the methods exist but waste-specific deployment does not (Sections 5.3–5.4), before appraising the tier (Section 5.5).

5.1 Generative families and the centrality of constraint

Three architectural families underpin Tier III, each with a different generative target. **Generative adversarial networks (GANs)** pit a generator against a discriminator to synthesize samples indistinguishable from real data, and have been the workhorse for realistic image synthesis [8]. **Diffusion models** generate by learning to reverse a gradual noising process and now define the state of the art in controllable, high-fidelity image and, increasingly, multi-modal synthesis; latent diffusion in particular made high-resolution generation computationally tractable, and text-to-image diffusion has become a general-purpose content-creation substrate [9,21,22]. **Transformer-based generators**—the autoregressive language and code models of Tiers II and IV—are themselves generative systems when their output is the artifact of interest: an LLM that drafts a compliance report or emits process-control code is performing Tier III generation.

What unifies the tier, and what makes it engineering-relevant rather than merely impressive, is constraint. Unconstrained generation produces plausible content; useful generation produces content that is faithful to physical, regulatory, or operational specifications. The defining methodological challenge of Tier III in USWM is therefore conditioning and validation: ensuring that a synthetic waste image preserves the discriminative features a downstream classifier must learn, that a generated process scenario respects mass-and-energy balances, and that an auto-drafted report is factually grounded rather than fluent fabrication. This constraint problem is the generative counterpart of the hallucination problem of Tier II, and it recurs as a cross-cutting reliability concern in Section 8.

5.2 The dominant application: generative data augmentation for perception

The one place where Tier III has produced concrete, evaluated USWM evidence is data augmentation in service of Tier I. Perception models are bottlenecked by the scarcity, cost, and class imbalance of annotated waste imagery (Sections 3.1–3.2), and generative synthesis offers a direct remedy: manufacture additional, varied, realistically labeled training samples for rare categories and difficult conditions. WasteGAN is the anchor exemplar—a GAN-based method that generates synthetic waste images together with their segmentation masks to augment training data for robotic waste sorting, with the evaluated outcome being improved downstream sorting performance [52]. Under our assignment protocol (Section 2.2), WasteGAN is a Tier III study precisely because its central, evaluated contribution is the generative method, even though its purpose is to serve perception; it is the clearest concrete instance of the perception-generation coupling foreshadowed in Section 3.2.

This single, well-evaluated study is also, tellingly, close to the entirety of the direct USWM generative literature, and its narrowness is informative. The application is instrumental (generation as a feeder for perception) rather than generative of engineering value in its own right; it operates on images, the modality the field already handles; and it inherits the central risk of synthetic augmentation—that a generator trained on scarce data may reproduce and amplify the very biases and gaps it is meant to fill, or introduce artifacts that degrade rather than improve the downstream model. That the tier’s most mature application is also its most instrumental signals how far USWM remains from exploiting generation for design, planning, and reporting.

5.3 The prospective frontier I: scenario, process, and design generation

Beyond data augmentation, the higher-value promise of Tier III is generative design: using generative models to propose, explore, and optimize waste-system configurations, treatment-process operating points, and infrastructure scenarios that a human engineer then evaluates. Diffusion and Transformer generators are capable, in principle, of producing candidate process designs, synthetic operating scenarios for stress-testing, and content to populate the simulation environments on which planning depends [9,21,22]. In USWM, however, this remains almost entirely prospective: peer-reviewed, evaluated applications of generative models to waste-process or waste-system design are essentially absent. The adjacent process-modeling literature that does exist—machine-learning prediction of waste-to-energy heating values, biogas yields, and emissions—is discriminative forecasting rather than generative design, and is treated under Tier IV (Section 6) as decision support rather than synthesis. The gap is thus not a lack of capable methods but a lack of waste-domain formulations that couple generative models to the physical and regulatory constraints a credible design must satisfy (Section 5.1).

5.4 The prospective frontier II: automated reporting and digital-twin content

A second prospective frontier is the generation of language and structured artifacts. LLMs can, in principle, automate the substantial documentary burden of waste management—compliance and monitoring reports, permit applications, public disclosures, and operational summaries—drafting them from structured facility data with citation-to-source grounding. This is Transformer-based generation (Section 5.1) applied to the documentary output of a waste system, and it is the natural extension of the comprehension capability of Tier II from answering to authoring. Relatedly, generative models can populate the content layer of **digital twins**—synthetic but realistic data streams, scenarios, and visualizations that animate a virtual replica of a facility for monitoring, training, and what-if analysis. Digital-twin methodology is beginning to enter waste operations [66], but the generative population of such twins, and the use of LLMs for grounded automated reporting in waste specifically, remain demonstrations-in-waiting rather than evidenced deployments. As in Section 5.3, the methods are mature elsewhere; their constrained, validated instantiation for waste is the missing step.

[Table 4 — placement: here, Section 5.4.] Tier III (Generate) in USWM: applications, methods, maturity, and the constraint/validation challenge. Columns: Application (data augmentation for perception; scenario/process/design generation; automated reporting; digital-twin content); Generative family (GAN / diffusion / Transformer-LLM); Direct USWM evidence (evaluated / demonstration / prospective); Governing constraint that must be enforced (label fidelity; mass-energy balance and regulatory feasibility; factual grounding; physical realism); Key risk (bias amplification and artifacts; infeasible designs; fabricated content; twin-reality divergence). The table makes explicit that only the first row is evidenced (WasteGAN [52]), substantiating the “early / exploratory” maturity rating of Table 1 and locating the white space the research agenda of Section 9 targets.

5.5 Appraisal of the Generate tier

Tier III is early and exploratory, and its profile is distinctive: of all five tiers, it exhibits the widest gap between the maturity of the general-purpose technique base and the poverty of its waste-domain application. Generation is mature and transformative in the broader AI landscape, yet in USWM it is represented almost solely by a single instrumental use—augmenting perception data [52]—with scenario design, automated reporting, and digital-twin content remaining prospective despite mature enabling methods.

Three observations carry forward. First, the tier’s binding obstacle is not capability but con-

straint enforcement and validation (Section 5.1): until generative outputs can be reliably conditioned on physical, regulatory, and operational specifications and validated against them, generation cannot move from producing training images to producing engineering-grade designs and documents. This is the same reliability theme that defines Tier II, now in synthetic rather than retrieval form, and it is consolidated in Section 8. Second, generation is doubly coupled to its neighbors: downward to Tier I, which it feeds with synthetic data, and upward to Tier IV, whose agentic systems must generate the plans, code, and tool-calls through which they act—so progress at Tier III is a precondition for the agentic capability of the next chapter. Third, the conspicuous absence of generative design and reporting in waste, against their maturity elsewhere, makes Tier III one of the highest-leverage opportunities the P2G map exposes: the methods are ready, the engineering need is acute, and the formulation work is largely undone.

[Figure 6 — placement: end of Section 5.] Schematic of the Tier III generative landscape in USWM, evidenced versus prospective. A two-zone diagram: a solid-bordered “evidenced” zone containing generative data augmentation for perception (GAN/diffusion → synthetic images + masks → improved Tier I sorting, anchored by WasteGAN [52]); and a dashed-bordered “prospective” zone containing scenario/process/design generation, LLM automated reporting, and generative digital-twin content. Each element is annotated with the governing constraint it must satisfy (label fidelity, mass-energy balance, factual grounding, physical realism). Downward and upward arrows connect the tier to Tier I (data feed) and Tier IV (plan/code generation for action), visually situating generation as the synthetic engine coupling perception to agency.

6. Tier IV — Act: Agentic Optimization and Control

Tier IV is where the P2G continuum crosses from advisory cognition to operational autonomy. The system here does not merely perceive, comprehend, or generate; it **plans and executes** sequences of actions toward a goal—typically by orchestrating tools, models, and controllers—closing the loop from perception and reasoning to the optimization and control of collection, sorting, and treatment operations. The human supervises rather than directs each step. This is the tier at which the agentic capabilities surveyed generally in Section 1.2 (tool use, multi-step planning, goal-directed autonomy [12,17,18]) would, if realized in USWM, most visibly depart from the conventional digitalization paradigm of Section 1.1. The chapter first separates two strands of evidence that the literature frequently elides: a substantial body of forecasting and prediction models that supply decision support but stop short of closing the action loop (Section 6.1), and a smaller body of optimization and control systems that do close it (Section 6.2). It then confronts directly the near-absence of genuinely agentic, LLM-orchestrated systems in USWM (Section 6.3), before appraising the tier (Section 6.4).

6.1 Forecasting and prediction: decision support short of agency

The most populous part of the Tier IV literature consists of deep-learning models that forecast waste-system variables for downstream human decision-making, without themselves acting. Generation forecasting is the paradigm case: LSTM-based and hybrid LSTM-metaheuristic models predict municipal solid waste (MSW) generation at city scale, supporting capacity planning [53,54], with tree-based ensemble methods reported as competitive alternatives for generation-rate forecasting in data-scarce municipal settings [77]; recurrent architectures model the seasonal and meteorological dependence of household disposal behavior [55]; and deep-learning forecasts of smart-bin fill status anticipate collection needs before bins overflow [56], with broader IoT-and-deep-learning smart-bin systems surveyed as an integrated sensing-and-prediction layer [57]. A parallel cluster targets the energy-recovery pathway: machine-learning models predict the higher heating value (HHV) of MSW from compositional

data to support waste-to-energy (WtE) feasibility assessment [61,62]; landfill methane generation estimates have been improved by machine-learning-optimized parameterization of the standard LandGEM model [63]; biogas production during anaerobic co-digestion has been forecast through engineered features and supervised learning [64], with explainable-ML approaches reviewed specifically for anaerobic digestion to make such predictions auditable to process engineers [65]; and a brain-inspired modular neural network has been used to predict NOx emissions in incineration processes for environmental-compliance monitoring [67].

These studies are appropriately classified at Tier IV under our protocol because their evaluated contribution is operational decision support for collection, treatment, and recovery systems rather than perception or generation—but it is essential to register that none of them closes the action loop. A forecast of bin fill level, HHV, or methane yield is handed to a human planner; the system does not itself dispatch a truck, adjust a feed rate, or schedule a maintenance action. They are, in the vocabulary of this review, the most mature available evidence for the lower half of Tier IV—optimization-relevant prediction—while the upper half, closed-loop control and autonomous planning, is far thinner.

6.2 Optimization and closed-loop control

A smaller set of studies does close the loop, either through classical optimization informed by learned models or through reinforcement learning (RL) that directly learns a control policy. On the logistics side, dynamic route optimization for IoT-enabled waste-collection vehicles adjusts collection routing in response to real-time bin and traffic data, automating a decision that forecasting alone leaves to a human dispatcher [58]; and deep reinforcement learning has been applied to the stochastic e-waste collection problem, learning a routing policy directly from simulated experience under uncertainty in waste arrivals, which is a genuine instance of the agentic, trial-and-error optimization that distinguishes RL from supervised forecasting [59]. On the process side, an AI-based control strategy has been developed for energy-efficient operation of a WtE incineration plant, adjusting process parameters to improve efficiency in a documented case study [60]; and digital-twin methodology has been implemented and field-validated to enhance production and service systems in waste management, providing the closed-loop simulation-and-feedback substrate on which model-based control and what-if planning depend [66].

These control- and optimization-oriented studies are the clearest direct evidence of Tier IV agency presently available in USWM, but they share a common character: the “agent” is a classical optimizer or an RL policy trained for a single, narrowly scoped control problem (a route, a feed-rate setpoint), not a general-purpose, tool-using, LLM-orchestrated agent of the kind now being developed elsewhere [12,17,18]. They demonstrate that closed-loop autonomy is achievable in waste operations; they do not yet demonstrate the large-model agentic paradigm.

6.3 The missing piece: LLM agents for waste operations

The general-purpose agentic paradigm—an LLM that reasons about a goal, selects and invokes external tools or sub-models, observes the result, and iterates, in the manner formalized by ReAct-style reasoning-and-acting loops [12] and surveyed broadly for autonomous LLM-based agents [17,18]—has, to our knowledge, no peer-reviewed, evaluated instantiation specific to waste-collection or waste-treatment operations. This is the most consequential evidence gap this review identifies above Tier I. The capability is not hypothetical: the same architectural pattern already orchestrates tool use in adjacent industrial domains, evidenced by emerging benchmarks for LLM-agent task automation in industrial asset operations and maintenance that test precisely the orchestration skills—tool selection, multi-step diagnosis, closed-loop verification—a waste-operations agent would require [80], and the component pieces required for a waste-operations agent largely exist piecemeal across the tiers already

surveyed—Tier I perception modules that could serve as an agent’s “eyes” on a sorting line, Tier II comprehension systems that could supply regulatory and procedural grounding, the forecasting models of Section 6.1 that could serve as callable sub-tools, and the RL and optimization controllers of Section 6.2 that could serve as the agent’s actuators. What is absent is the orchestrating layer that would compose these components into a goal-directed, LLM-supervised control loop for collection, sorting, or treatment, together with the safety architecture—bounded action spaces, human-in-the-loop override, verifiable tool outputs—that closing such a loop in a physical, safety-relevant system would require. We treat this absence not as a peripheral gap but as the single clearest research-agenda item the P2G map exposes at this tier (Section 9).

[Table 5 — placement: here, Section 6.3.] Tier IV (Act) evidence stratified by closed-loop status. Columns: Study (ref.); Operational target (generation forecasting / smart-bin monitoring / WtE-process prediction / route optimization / incineration control / digital twin); Method (LSTM / RNN / supervised ML / classical optimization / RL / AI control strategy); Loop status (open — forecast handed to human; closed — system adjusts an operational variable); Agentic orchestration present (none / RL policy / classical optimizer / LLM-agent — none observed). The table is constructed to make visually explicit that the “open-loop forecasting” column is densely populated while the “LLM-agent orchestration” column is empty, substantiating the “nascent” maturity rating of Table 1.

6.4 Appraisal of the Act tier

Tier IV is nascent. The evidence base, on inspection, decomposes into a mature stratum of open-loop forecasting and prediction (Section 6.1), a thinner but real stratum of closed-loop optimization and control via classical methods and reinforcement learning (Section 6.2), and an essentially empty stratum of general-purpose, LLM-orchestrated agentic operation (Section 6.3). Reporting these as a single undifferentiated “AI for waste operations” literature—as a life-cycle-stage synthesis would tend to do, since all three sit within the “collection” or “treatment” stage—would obscure exactly the gradient that matters: the field has solved prediction, partially solved control, and not yet attempted agency.

Two implications follow for the remainder of the review. First, the components for a waste-operations LLM agent already exist, scattered across Tiers I-III and the lower half of Tier IV; the integration problem—not basic capability—is what currently blocks Tier IV from advancing, making this tier a question of systems engineering and safety architecture as much as of model capability. Second, the risk profile of closing this loop is categorically higher than at any tier surveyed so far: an agent that mis-acts on a misperceived input or a hallucinated regulatory interpretation propagates the failure modes of Tiers I-III into physical consequences—misrouted vehicles, damaged equipment, unsafe process excursions—which is why human supervision, verifiable tool outputs, and bounded autonomy are not optional design choices but preconditions for responsible Tier IV deployment, a theme consolidated with the tier’s other cross-cutting risks in Section 8.

[Figure 7 — placement: end of Section 6.] Schematic of the Tier IV action stack and the agentic integration gap. A loop diagram showing, around a central “goal” node, four component blocks—Perceive (Tier I sensing), Comprehend (Tier II grounding), Generate (Tier III synthetic planning content)—feeding into a “Decide & Act” block that currently bifurcates into two solid-line paths (open-loop forecasting → human decision; closed-loop classical/RL control → operational adjustment) and one dashed, unrealized path labeled “LLM-agent orchestration (ReAct-style),” annotated with the safety-architecture requirements (bounded action space, human override, verifiable tool outputs) that would need to accompany its closure.

7. Tier V — Govern: Human-AI Symbiosis and Governance

The summit of the P2G continuum reframes the central question of the review. Tiers I-IV ask what a large model can do to the waste stream—perceive, comprehend, generate, act. Tier V asks how machines, citizens, and institutions should jointly govern the waste system, and how the AI itself should be governed in turn. The system at this tier participates in the social and institutional regulation of waste: mediating public engagement and behavior, supporting policy co-design, and—reflexively—surfacing the questions of trust, accountability, equity, and environmental footprint that large-model deployment raises. “Autonomy” is here reframed as shared agency: the defining demand is value-laden judgment, and the appropriate object of analysis is the socio-technical system rather than the model in isolation. This is also the tier whose very existence the conventional life-cycle ontology is structurally unable to register (Section 1.3): governance belongs to no waste stage, and is therefore the clearest demonstration of why a technology-native framework is required. The chapter treats the two faces of governance in turn—AI as an instrument of waste governance (Section 7.1) and AI as an object of governance (Sections 7.2–7.3)—and then appraises the tier and, with it, the upper boundary of the continuum (Section 7.4).

7.1 AI as an instrument of governance: engagement, behavior, and policy co-design

The forward-facing role of Tier V is to use large models to strengthen the social machinery of waste management—citizen participation, behavioral change, and participatory policy design, a role for which the broader case for ChatGPT-class models as aids to environmental policymaking has already been articulated in principle [79]. Conversational and language interfaces can lower the barrier to public engagement: answering residents’ sorting and disposal questions in natural language, delivering targeted educational nudges, and channeling community input into planning. Direct USWM evidence for this role is, however, almost entirely prospective. The most relevant peer-reviewed exemplar comes from the adjacent urban-environmental domain: large language models have been examined for enhancing citizen engagement in urban greening through value co-creation, demonstrating the mechanism by which an LLM interface can mediate between residents and an environmental-management institution [74]. The transfer to waste—LLM-mediated citizen engagement in source separation, participatory siting of facilities, or co-design of collection schemes—is immediate in principle and unevicenced in practice. The water sector’s deployed conversational agents for public communication and education [28] supply the closest demonstration that such interfaces can be operationalized, reinforcing the leading-indicator pattern observed at Tier II. The behavioral channel—using generative interfaces to influence disposal and consumption behavior—raises its own governance questions of manipulation, consent, and equity that belong squarely to the reflexive half of this tier, to which we now turn.

7.2 AI as an object of governance I: the environmental footprint and the AI sustainability paradox

The most environmental-engineering-specific contribution of Tier V, and the one most thoroughly occluded by a life-cycle organization, is the recognition that the large models deployed across Tiers I-IV carry a material environmental footprint of their own. Training and serving large models consume substantial energy and water and emit carbon: foundational analyses quantified the energy and carbon cost of training large NLP models [70] and of large neural-network training in data centers [72], motivated the “Green AI” program that treats computational efficiency as a first-class research value alongside accuracy [71], and—through the “Anatomy of an AI System” analysis—traced the full material and extractive supply chain behind a single AI service [73].

For a field whose *raison d’être* is environmental protection, this creates a distinctive and unavoidable tension, which we term the **AI sustainability paradox**: the same large models

deployed to improve waste sorting, divert material from landfill, optimize collection routes, and reduce emissions themselves consume energy, water, and materials and generate electronic waste at end of life—indeed feeding back into the very waste stream the field manages. A perception model that marginally improves recycling purity may, if it is large and frequently retrained, carry a net environmental cost that exceeds its benefit; the question is empirical and system-dependent, and it is one that the USWM literature has scarcely begun to pose, let alone answer with life-cycle accounting. No life-cycle-stage framework can surface this paradox, because the AI’s footprint is not located at any stage of the waste flow; only a technology-native framework that makes the model itself an object of analysis can. We therefore treat the paradox as a central cross-cutting finding, developed quantitatively as a research need in Section 8 and as an agenda item in Section 9.

7.3 AI as an object of governance II: trust, accountability, equity, and regulation

The second reflexive concern is the trustworthy and equitable governance of AI in waste systems. The foundational critiques of large language models apply directly: the “stochastic parrots” analysis warns of the environmental and social costs and the embedded biases of ever-larger models [68], and systematic taxonomies of the ethical and social risks of language models catalogue harms—misinformation, discrimination, and the concentration of power—that any deployment must anticipate [69]. These concerns acquire specific force in USWM because of its acute environmental-justice dimension (Section 1.1): waste infrastructure burdens fall disproportionately on vulnerable communities, and an AI system that allocates collection resources, sites facilities, or targets enforcement can either mitigate or entrench that inequity depending on how its objectives, training data, and error distribution are governed. Accountability for automated waste decisions, transparency of model reasoning to affected residents, and the equitable distribution of benefits and harms are therefore not generic AI-ethics boilerplate but domain-critical design constraints.

This governance is increasingly being formalized in hard law. The European Union’s Artificial Intelligence Act establishes a risk-tiered regulatory regime for AI systems [75], and AI systems that inform resource allocation, environmental compliance, or facility siting may fall within its higher-risk categories, with attendant obligations of transparency, human oversight, and conformity assessment. The regulatory environment is thus moving faster than the peer-reviewed USWM-AI literature, which has scarcely engaged with the compliance, documentation, and oversight requirements that deployment will entail—a gap between technical research and governance reality that is itself a Tier V finding.

[Table 6 — placement: here, Section 7.3.] Tier V (Govern) concerns, mechanisms, and evidence status. Two-part structure. Part A — AI as instrument: concern (citizen engagement, behavioral nudging, participatory policy co-design); mechanism (conversational/LLM interface); USWM evidence status (prospective; nearest exemplars in urban greening [74] and water-sector public communication [28]). Part B — AI as object: concern (environmental footprint / sustainability paradox; bias and equity; accountability and transparency; legal compliance); grounding literature ([70,71,72,73]; [68,69]; [68,69]; [75]); USWM-specific engagement (largely absent — a research gap). The table makes explicit that Tier V is almost entirely prospective in waste and that its grounding is imported from general AI-governance and adjacent-domain sources, substantiating the “largely prospective” maturity rating of Table 1.

7.4 Appraisal of the Govern tier

Tier V is largely prospective in USWM: there is, at present, essentially no peer-reviewed work that deploys and evaluates large models for waste governance, whether as instrument

(engagement, policy co-design) or as the explicit object of footprint, equity, and accountability analysis within the waste domain. Its content in this review is therefore necessarily constructed by importing the general AI-governance literature [68–73,75] and the nearest adjacent-domain exemplars [28,74] and projecting their implications onto waste—an honest reflection of the state of the field rather than a limitation of the framework.

Three closing observations elevate this from an admission of sparsity to a substantive finding. First, the emptiness of Tier V is the strongest vindication of the P2G framework’s central claim: the governance concerns that matter most for the responsible deployment of large models—footprint, trust, equity, accountability—are precisely the ones a life-cycle-stage review cannot see, and their systematic absence from the existing literature is plausibly a consequence of the prevailing ontology’s blind spot, not merely a coincidence of immaturity. Second, the AI sustainability paradox (Section 7.2) is, for an environmental-engineering audience, the single most important issue this tier raises, and it is methodologically tractable: it calls for life-cycle assessment of AI-enabled waste interventions that nets the model’s own footprint against its environmental benefit, a study design the field is fully equipped to execute and has not. Third, governance is not the optional capstone of the continuum but its enabling condition: the higher tiers cannot be responsibly ascended without it, since the autonomy of Tier IV and the synthetic authority of Tiers II–III become socially acceptable only inside a trust, accountability, and oversight architecture that Tier V defines. The cross-cutting analysis of Section 8 now consolidates these governance concerns with the reliability, data, and standardization issues surfaced tier by tier.

[Figure 8 — placement: end of Section 7.] Schematic of the two faces of Tier V governance. A balance/feedback diagram with the waste socio-technical system at the center. On the left, “AI as instrument”: LLM interfaces connecting citizens, operators, and institutions for engagement, nudging, and policy co-design (solid where evidenced in adjacent domains [28,74], dashed for waste). On the right, “AI as object”: a stack of governance concerns—environmental footprint, bias/equity, accountability/transparency, legal compliance ([70–73]; [68,69]; [75])—with a prominent feedback arrow from the deployed AI of Tiers I–IV back into the waste stream (electronic waste, energy, water), visually encoding the AI sustainability paradox. A surrounding “human–AI shared-agency” boundary frames the whole, marking governance as the condition within which the lower tiers operate.

8. Cross-Cutting Maturity-and-Gap Analysis and Critical Appraisal

The five preceding chapters synthesized the evidence tier by tier. This chapter steps back and reads the tiers together, for two purposes that a tier-by-tier narrative cannot itself accomplish. First, Section 8.1 consolidates the maturity gradient previewed in Table 1 into an explicit map of where the field’s effort is concentrated and where it is absent—the central empirical finding the P2G framework was built to expose. Second, Sections 8.2–8.6 identify the issues that recur across every tier rather than belonging to any one of them: reliability and hallucination, data and infrastructure scarcity, the absence of standardization and benchmarks, equity and environmental justice, and the AI sustainability paradox introduced in Chapter 7. These cross-cutting issues are, we argue, precisely the class of finding that the conventional life-cycle-stage ontology is structurally unable to surface, because each cuts across every stage of the material flow while belonging to none of them (Section 1.3). Section 8.7 closes with a critical appraisal of the review itself.

8.1 The maturity gradient, consolidated

Reading Chapters 3–7 in sequence traces a single, monotonic gradient of evidentiary density that falls as cognitive sophistication and operational autonomy rise. Tier I (Perceive) is established and consolidating, with a dense and still-growing literature spanning closed-set

CNN/ViT classification, real-time YOLO detection, and—genuinely large-model—open-vocabulary VLM recognition and promptable segmentation [39–51]. Tier II (Comprehend) is emerging: the enabling methods are mature, but direct, benchmarked USWM applications are scarce, with the adjacent water sector standing as the field’s clearest leading indicator [25–29]. Tier III (Generate) is early and exploratory, represented in USWM almost solely by one instrumental application—generative augmentation of perception data [52]—while scenario design, automated reporting, and digital-twin content remain prospective despite mature general-purpose methods. Tier IV (Act) is nascent: open-loop forecasting is well evidenced [53–57,61–65,67], closed-loop optimization and control exist but are narrow and non-agentic [58–60,66], and general-purpose LLM-orchestrated agentic operation has, to our knowledge, no peer-reviewed USWM instantiation. Tier V (Govern) is largely prospective, its content in this review necessarily constructed by importing general AI-governance literature and adjacent-domain exemplars [28,68–75] rather than drawn from evaluated waste-domain deployments.

This gradient is, we submit, the central empirical finding of the review, and its significance lies as much in why it is visible as in its content. Under the conventional life-cycle-stage ontology, the abundant perception work of Tier I is distributed across every stage—generation monitoring, collection-point sorting, treatment-line classification, recovery-facility inspection—so that each stage chapter appears comparably well populated, and the reader receives an impression of broadly uniform technological maturity across the waste system. The P2G reclassification removes that distribution and concentrates the same perception studies into a single tier, which simultaneously reveals how much of the field’s apparent activity is doing the same cognitive work and how starkly the higher-order capabilities are missing once they are no longer diluted across stage chapters. The gradient is not a new fact about the literature; it is a fact that only becomes visible once the literature is organized by the principle this review proposes.

[Table 7 — placement: here, Section 8.1.] Consolidated maturity-and-gap map across the five P2G tiers. Columns: Tier; Maturity (from Table 1); Approximate evidence density (qualitative: dense / moderate / thin / minimal / near-absent); Dominant evidence type (e.g., closed-set classifiers and VLMs; advisory/embedded comprehension; instrumental data augmentation; open-loop forecasting; imported general-AI governance literature); Principal white space (e.g., operational-reliability benchmarking; waste-domain RAG corpora and benchmarks; generative process/report design; LLM-agent orchestration and safety architecture; evaluated waste-governance deployments). This table is the quantitative-in-spirit, qualitative-in-execution summary of Section 8.1, intended as the single reference figure a reader returns to for the review’s headline finding.

[Figure 9 — placement: end of Section 8.1.] The P2G maturity gradient, rendered as a descending step or “funnel” diagram. Five bars or steps, one per tier, with qualitative height/width encoding evidence density (tallest at Perceive, shortest at Govern), annotated with the dominant evidence type and principal white space drawn from Table 7. A faint overlay of the conventional life-cycle-stage axis (generation → collection → sorting/treatment → recovery → disposal) running orthogonally beneath the funnel visually reprises Figure 1 and makes the contrast concrete: the same studies that look “evenly spread” along the stage axis collapse into a single tall step when read along the cognition/autonomy axis.

8.2 Reliability and hallucination as a continuum-wide property

Every tier above Perceive exhibits a version of the same underlying risk, even though each chapter named it differently: visible misclassification at Tier I gives way to fluent failure modes that are harder to detect specifically because the output is well-formed. Tier II’s

central risk is hallucinated or unsupported regulatory and technical claims, which RAG and citation-to-source exist to mitigate but which remain unvalidated for waste-specific corpora (Section 4.4) [10]. Tier III’s constraint-enforcement problem—ensuring generated artifacts respect physical, regulatory, or label-fidelity constraints—is hallucination’s synthetic-generation analogue (Section 5.1,5.5). Tier IV compounds both: an agent that acts on a misperceived input or a hallucinated regulatory interpretation converts an epistemic failure into a physical one (Section 6.4). Tier V governance instruments exist specifically because the social and institutional costs of these failures—misinformation, biased allocation, unaccountable automated decisions—do not stay contained within a single tier [68,69]. Treated tier-by-tier, reliability looks like five separate caveats; treated cross-cuttingly, it is a single design requirement—verifiable grounding, bounded autonomy, and human oversight—that must be engineered into every tier above Perceive before deployment, and whose absence at any one tier propagates upward.

8.3 Data, compute, and infrastructure scarcity

A second recurring constraint is the scarcity of waste-domain resources needed to adapt general-purpose large models reliably. At Tier I, labeled image scarcity motivated the generative-augmentation literature of Tier III [39,40,52]; at Tier II, the absence of curated, authoritative waste-regulation and operational-record corpora is the immediate bottleneck to RAG-grounded comprehension, exactly where the water sector advanced by building domain-adapted models and benchmarks [26,27]; at Tier III, the lack of waste-specific formulations that encode physical and regulatory constraints blocks generative design and reporting (Section 5.3–5.4); at Tier IV, safe agentic deployment requires verifiable simulation and testing environments that do not yet exist for waste operations (Section 6.3). The pattern across tiers is consistent: the general-purpose model and method are typically already available; what is missing is the curated, domain-specific data, benchmark, or simulation infrastructure required to adapt and validate it for waste. This reframes “more research” as, more specifically, “more domain infrastructure”—a sharper and more actionable diagnosis than the literature’s tier-by-tier framing yields, and one taken up directly in the research agenda of Section 9.

8.4 Absence of standardization and benchmarks

Closely related to data scarcity is the near-total absence, across Tiers II–V, of standardized benchmarks against which competing systems and claims can be compared. Tier I has converged on shared benchmarks (TACO, WaRP, and the emerging Waste-Bench for VLMs) that enable cumulative, comparable progress [39,40,46]; no analogous benchmark exists for waste-knowledge comprehension, generative-design fidelity, agentic task completion, or governance-instrument effectiveness. The water sector’s water-knowledge benchmark [27] again supplies the template USWM lacks. Without such instruments, claims of capability at Tiers II–V are difficult to compare across studies or to validate before deployment, which is itself a barrier to the field’s maturation independent of any single tier’s underlying model capability.

8.5 Equity and environmental justice

The environmental-justice dimension flagged at Tier V (Section 7.3) is, on reflection, not confined to governance; it is latent in every tier’s deployment decisions. Perception systems trained predominantly on imagery from well-resourced facilities may underperform in under-resourced ones (Tier I); comprehension systems grounded in formally codified regulation may be less useful where waste management is informal (Tier II); forecasting and optimization systems calibrated on data-rich cities may misallocate collection resources in data-poor ones

(Tier IV); and only Tier V explicitly names accountability and equity as design objectives (Section 7.3) [68,69]. Equity is therefore best understood not as a Tier V topic but as a cross-cutting evaluation criterion that the field has, so far, applied almost exclusively at the top of the continuum and almost nowhere below it.

8.6 The AI sustainability paradox, revisited

Chapter 7 introduced the AI sustainability paradox—the tension between the environmental benefit large models deliver and the environmental cost of producing that benefit [70–73]—as a Tier V finding. Read cross-cuttingly, however, the paradox is generated at every tier and merely recognized at Tier V: every perception model retrained for a new facility, every comprehension system queried at scale, every generative-augmentation run, and every agentic control loop consumes energy and, indirectly, materials. The paradox is therefore not a governance add-on but an accounting problem latent in Tiers I–IV that only Tier V’s reflexive stance makes visible. This is, in our reading, the most distinctive substantive contribution that an environmental-engineering audience should take from this review: a life-cycle assessment of AI-enabled USWM interventions—netting model footprint against operational and environmental benefit—is both overdue and, given the field’s existing LCA expertise, immediately executable.

8.7 Critical appraisal of the review

Honesty about this review’s own limitations is itself part of the cross-cutting analysis it argues the field needs. Three limitations are material. First, the evidentiary asymmetry that the P2G map documents is partly a function of true field immaturity and partly a function of search recency and disciplinary visibility—Tier IV and V capabilities may exist in grey literature, industry deployments, or adjacent disciplines not indexed by the databases searched (Section 2.3), and the “near-absent” characterization of upper-tier USWM evidence should be read as “not found in the peer-reviewed and preprint literature surveyed,” not as a categorical claim of nonexistence. Second, the qualitative, narrative synthesis adopted in place of a quantified PRISMA flow (Section 2.3) trades precision for transparency about an immature, fast-moving corpus; a future quantitative scientometric study, explicitly flagged as out of scope here, would sharpen the density estimates offered qualitatively in Section 8.1. Third, the tier-assignment protocol (Section 2.2), while reproducible in principle, requires judgment calls for genuinely multi-tier studies, and a different reasonable assignment of a handful of boundary cases (e.g., VLM perception studies that lean heavily on language comprehension) would shift but not overturn the gradient documented here. None of these limitations weakens the central claim that a technology-native organizing framework reveals structure the life-cycle ontology cannot; they bound the precision, not the direction, of the finding. The next chapter converts this appraisal into a structured, tier-organized research agenda.

9. Challenges and Future Research Directions

The maturity-and-gap analysis of Chapter 8 identified what is missing; this chapter converts that diagnosis into a structured agenda for what should be built next. The agenda is organized in two layers, consistent with the architecture of the review as a whole. Sections 9.1–9.5 set out tier-specific priorities, each keyed to the principal white space identified for that tier in Table 7. Section 9.6 then sets out priorities for the cross-cutting issues of Chapter 8, which by construction cut across every tier and therefore cannot be discharged by progress at any single one. Section 9.7 closes by proposing a small set of flagship integrative studies through which the field could demonstrate, in a single evaluated system, the climb from perception to governance that this review has so far only mapped.

9.1 Tier I (Perceive): from closed-set saturation to operational robustness

Tier I is dense but, as Section 3.5 argued, its saturation is largely a saturation of pre-foundation-model methods; the genuinely large-model capability—open-vocabulary and zero-/few-shot VLM recognition—is recent and not yet operationally robust. The priority is therefore not more classifiers but **rigorous robustness evaluation of foundation-model perception under realistic operating conditions**: systematic benchmarking of VLM and promptable-segmentation performance across facilities, lighting, occlusion, and contamination regimes, building on and extending Waste-Bench-style infrastructure [46]. A second priority is **multimodal extension beyond the visible spectrum**: foundation models that natively fuse RGB imagery with NIR/multispectral and other sensor streams remain largely absent (Section 3.4), and would unify currently separate spectral and visual perception literatures [50]. A third, more modest but high-value priority is **standardized reporting of failure cases**, not only aggregate accuracy, so that downstream tiers (especially Tier IV) can be designed around known, characterized perception failure modes rather than an assumed near-perfect perception layer.

9.2 Tier II (Comprehend): waste-domain grounding and benchmarking

Tier II’s bottleneck, as Section 4.4 established, is the absence of curated waste-domain knowledge corpora and evaluation benchmarks against which RAG-grounded comprehension can be built and validated. The single highest-leverage action item is therefore the construction of an **authoritative, machine-readable corpus of waste regulations, standards, and operational records**, paralleling the domain-adapted models and benchmarks that advanced the water sector [26,27], together with a **waste-knowledge benchmark** that quantitatively evaluates regulatory interpretation, technical question answering, and decision-support accuracy rather than relying on anecdotal demonstration. A second priority is **systematic evaluation of hallucination rates and citation fidelity** for RAG-grounded waste-comprehension systems specifically, since the compliance-critical nature of waste regulation makes silent hallucination a higher-stakes failure than in many other LLM application domains. A third is **knowledge-graph construction for the waste domain**, structuring the entities and relations (materials, regulations, facilities, permits) that would make retrieval and reasoning more precise and auditable than unstructured RAG alone.

9.3 Tier III (Generate): from instrumental augmentation to constrained design

Tier III’s defining gap, per Section 5.5, is the chasm between mature general-purpose generative methods and their almost entirely instrumental waste-domain application (data augmentation for perception). The priority research direction is **constraint-aware generative design**: formulating generative methods for waste-process and waste-system design that explicitly encode and validate against physical (mass-energy balance), regulatory, and operational feasibility constraints, rather than optimizing for visual plausibility alone (Section 5.1). A second direction is **grounded automated reporting**: LLM systems that draft compliance and operational documentation with verifiable citation to underlying facility data, extending Tier II’s grounding requirement into the generative register (Section 5.4). A third is the **generative population of digital twins** with synthetic but physically realistic operational data and scenarios, building on emerging digital-twin methodology in waste operations [66] to support training, monitoring, and what-if analysis without requiring exhaustive real-world data collection.

9.4 Tier IV (Act): integration and safety architecture for agentic operation

Tier IV’s central finding (Section 6.4) is that the components of a waste-operations agent already exist piecemeal—perception, comprehension, generation, forecasting, and narrow

optimization and control—while the orchestrating, agentic layer that would compose them does not. The foremost research priority is therefore **the design and evaluation of LLM-orchestrated agentic systems for bounded waste-operations tasks**, beginning with low-stakes, reversible actions (e.g., route re-planning recommendations subject to dispatcher approval) before progressing toward higher-autonomy, higher-consequence control (e.g., direct process-parameter adjustment), in the manner pioneered by ReAct-style reasoning-and-acting architectures [12] but with a safety architecture purpose-built for physical operations. That architecture must specify, and be evaluated against, **bounded action spaces, verifiable tool outputs, and human-override mechanisms** as first-class design requirements rather than post-hoc safeguards. A second priority is the development of **high-fidelity simulation and digital-twin environments** in which agentic policies can be trained and stress-tested before any deployment touches physical infrastructure, extending [66] toward agent-training use. A third is **closing the loop between forecasting and control**: the abundant open-loop forecasting literature (Section 6.1) is a substantial existing asset that has not yet been coupled to the optimization and RL methods of Section 6.2 into integrated closed-loop systems.

9.5 Tier V (Govern): from imported principles to evaluated instruments

Tier V’s content in this review was necessarily constructed by importing general AI-governance principles and adjacent-domain exemplars (Section 7.4); the research priority is to replace import with evaluated, waste-domain-specific instruments. This includes **piloted LLM-mediated citizen-engagement systems for waste behaviors**—source separation, illegal-dumping reporting, participatory facility siting—evaluated for actual behavioral and participatory effect rather than asserted by analogy to adjacent domains [28,74]. It includes **life-cycle assessment of AI-enabled waste interventions** that explicitly nets the energy, water, and material footprint of the deployed models (Tiers I-IV) against their operational and environmental benefit, operationalizing the AI sustainability paradox identified in Sections 7.2 and 8.6 as a quantitative LCA research program rather than a qualitative caveat. It includes **equity-stratified evaluation protocols** that assess perception, comprehension, forecasting, and agentic systems across resource-rich and resource-poor deployment contexts, converting the cross-cutting equity concern of Section 8.5 into a standard evaluation requirement rather than a Tier V afterthought. Finally, it includes **applied compliance research** mapping specific USWM AI use cases against emerging regulatory regimes such as the EU AI Act [75], to close the gap identified in Section 7.3 between the pace of regulation and the pace of peer-reviewed engagement with it.

[Table 8 — placement: here, end of Section 9.5.] Tier-organized research agenda. Columns: Tier; Principal white space (from Table 7); Priority research direction(s); Required infrastructure (benchmark / corpus / simulation environment / evaluation protocol); Illustrative near-term deliverable (e.g., a VLM robustness benchmark; a waste-regulation RAG corpus and QA benchmark; a constraint-validated generative-design proof of concept; a bounded-autonomy agent pilot for route re-planning; a footprint-netted LCA of a deployed perception system). The table is designed as a planning reference: each row names a concrete, scopable project rather than a general aspiration, so that the agenda can be read as a set of fundable, evaluable research proposals rather than a list of open questions.

9.6 Cross-cutting priorities

Because the issues of Chapter 8 cut across tiers, progress at any single tier cannot resolve them; they require dedicated, cross-cutting research investment. **Reliability and hallucination** (Section 8.2) call for a unified evaluation framework—applicable across Tiers II-IV—that measures grounding fidelity, constraint satisfaction, and failure-mode propagation

rather than tier-specific accuracy alone. **Data and infrastructure scarcity** (Section 8.3) calls for coordinated, ideally consortium-level investment in shared waste-domain corpora, benchmarks, and simulation environments, since the fixed cost of building such infrastructure is more efficiently borne once across the field than repeatedly within individual studies. **Standardization** (Section 8.4) calls for the research community to converge on benchmark suites for Tiers II-V analogous to those Tier I already enjoys, ideally coordinated through a community challenge or shared leaderboard. **Equity and environmental justice** (Section 8.5) calls for equity-stratified evaluation to become a default reporting requirement across all tiers, not a Tier V specialty. **The AI sustainability paradox** (Section 8.6) calls for standardized footprint reporting (energy, water, embodied carbon, end-of-life material burden) to accompany capability claims for any large model proposed for USWM deployment, enabling the netted LCA comparisons that Section 9.5 identifies as overdue.

9.7 Flagship integrative studies

The tier-specific and cross-cutting priorities above are best pursued, ultimately, not only as isolated projects but as components of a small number of **flagship integrative studies** that would demonstrate the P2G continuum operating end to end on a single, real waste system. One plausible flagship is a **fully instrumented sorting facility** in which a VLM perception layer (Tier I), a RAG-grounded compliance-and-operations comprehension layer (Tier II), a generative-augmentation and synthetic-reporting layer (Tier III), and a bounded-autonomy agentic control layer (Tier IV) are integrated and jointly evaluated, with an accompanying footprint and equity assessment (Tier V) reported alongside operational performance. A second is a **municipal collection system** in which forecasting, route optimization, and a citizen-engagement conversational interface are integrated and evaluated for combined operational and participatory outcomes. Such integrative studies would do for the upper tiers what the perception literature has already done for Tier I: convert a fragmented set of component demonstrations into a benchmarked, reproducible body of evidence, and they would constitute the most direct possible test of the central premise of this review—that large models reconfigure the relationship between human expertise and machine capability across the whole waste system, not merely at any one stage of it.

10. Conclusions

This review set out from a structural observation: that well over 90% of existing reviews of artificial intelligence in urban solid waste management (USWM) organize the literature by where in the material flow a technology operates, and that this life-cycle-stage ontology, however productive for the task-specific machine learning it was built to describe, cannot represent what large models and generative AI actually do, because their defining property is a level of cognition and autonomy that is stage-agnostic by construction (Sections 1.1–1.3). In response, we proposed and developed the **Perception-to-Governance (P2G) capability framework**, an original, technology-native ordering of the literature along a single ascending axis of machine cognition and autonomy—Perceive, Comprehend, Generate, Act, Govern (Section 2.1)—together with an explicit, reproducible tier-assignment protocol (Section 2.2) and a qualitative literature search and synthesis methodology suited to a fast-moving, still-immature corpus (Section 2.3).

Applying this framework to the literature yielded the review’s central empirical finding: a steep and consistent maturity gradient. Tier I (Perceive) is established and consolidating, with vision and increasingly vision-language models delivering robust, if not yet fully operationally validated, multimodal waste characterization (Chapter 3). Tier II (Comprehend) is emerging, its enabling methods mature but its waste-specific, benchmarked applications scarce, with the water and wastewater sector standing as the clearest leading indicator of what domain-adapted, RAG-grounded comprehension could become in waste (Chapter 4). Tier III (Gener-

ate) is early and exploratory, represented in USWM almost entirely by one instrumental use—generative augmentation of perception training data—while generative design, automated reporting, and digital-twin content remain prospective despite mature general-purpose methods (Chapter 5). Tier IV (Act) is nascent: forecasting is well evidenced, narrow closed-loop optimization and control exist, but general-purpose, LLM-orchestrated agentic operation of waste systems has, to our knowledge, no peer-reviewed instantiation (Chapter 6). Tier V (Govern) is largely prospective, its treatment here necessarily built by importing general AI-governance principles and adjacent-domain exemplars rather than evaluated waste-domain deployments (Chapter 7).

Reading these tiers together (Chapter 8) showed that this gradient is invisible under a life-cycle-stage organization, which distributes the abundant Tier I work across every stage and thereby manufactures an impression of broadly uniform technological maturity that the evidence does not support. It also surfaced issues that recur across every tier and belong to none of them in a stage-based reading: a continuum-wide reliability and hallucination problem that shifts in character but not in kind as autonomy rises; a domain-infrastructure scarcity that recurs identically at every tier above perception; a near-total absence of standardization and benchmarks above Tier I; an equity and environmental-justice dimension currently engaged almost exclusively at Tier V despite being latent throughout; and, distinctively for an environmental-engineering readership, the **AI sustainability paradox**—the recognition that the large models deployed to improve waste outcomes carry their own non-trivial energy, water, material, and end-of-life footprint, an accounting problem the field is equipped to address and has not yet undertaken. Chapter 9 converted this diagnosis into a structured, tier-organized research agenda, culminating in a proposal for flagship integrative studies that would demonstrate the full Perceive-to-Govern continuum operating end to end on a single real waste system.

Three implications follow for the field. First, the maturity gradient documented here is not a verdict on the value of large models in USWM but a map of where research effort has, and has not yet, gone; the gap at Tiers III–V reflects the field’s youth more than any inherent limit on what generative AI could contribute to waste generation, sorting, treatment, recovery, or disposal once the data, benchmark, and safety infrastructure identified in Chapter 9 are built. Second, the white space is itself informative: it is concentrated precisely where cognition and autonomy are highest, suggesting that the field’s near-term tractable advances likely lie in Tier II comprehension—where the enabling methods are most mature relative to their waste-domain application—while genuinely autonomous Tier IV operation and Tier V governance will require longer-horizon investment in safety architecture, simulation infrastructure, and evaluated social deployment. Third, and most generally, the exercise of reclassifying a substantial literature along a cognition-and-autonomy axis rather than a material-flow axis demonstrates that organizing principle is not a neutral expository choice; it determines which patterns in a body of evidence become visible and which remain hidden. We expect the P2G framework, or organizing logics in its spirit, to be of continuing use as large models diffuse further into USWM and into adjacent environmental-engineering domains that face the same mismatch between a physical-process ontology and a technology whose defining variation is cognitive rather than spatial.

The urban solid waste predicament described at the outset of this review (Section 1.1) will not be resolved by perception alone, however mature that capability becomes. It will require the field to climb the full continuum this review has mapped—from perceiving the waste stream, to comprehending the knowledge that governs it, to generating the artifacts that plan and document its management, to acting upon it with appropriate autonomy and safeguards, and ultimately to governing, with citizens and institutions, the human–AI systems on which that management increasingly depends. We hope the Perception-to-Governance framework offered here provides both a clearer picture of how far that climb has progressed and a usable map for completing it.

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